

# Homotopy Theory

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# 1 Simplicial Sets

## 1.1 Simplicial complexes

Recall that a **simplex** in  $\mathbb{R}^N$  is a subspace,

$$[v_0, \dots, v_n] = \left\{ \sum_i \lambda_i v_i \mid \sum_i \lambda_i = 1, 0 \leq \lambda_i \in \mathbb{R} \right\},$$

where  $v_i \in \mathbb{R}^N$  are **geometrically independent** (if  $\sum_i \lambda_i v_i = 0$  and  $\sum_i \lambda_i = 0$  then  $\lambda_i = 0$ ). The **faces** of a simplex  $[v_0, \dots, v_n]$  are the simplices generated by a subset of  $\{v_0, \dots, v_n\}$ . The **standard  $n$ -simplex** is the one generated by the standard basis vectors of  $\mathbb{R}^{n+1}$ :

$$|\Delta^n| = [e_0, \dots, e_n] = \left\{ (x_0, \dots, x_n) \in \mathbb{R}^{n+1} \mid \sum_i x_i = 1, x_i \geq 0 \right\} \subseteq \mathbb{R}^{n+1}.$$

A **simplicial complex** in  $\mathbb{R}^N$  is a collection of simplices  $K$  such that the face of any simplex in  $K$  is also in  $K$ , and the intersection of any two simplices in  $K$  is a face of both of them. Given a simplicial complex  $K$ , we can define its **geometric realization**  $|K|$  to be the union of all the simplices in  $K$ . A subset  $F \subseteq |K|$  is closed iff  $F \cap \sigma$  is closed in the simplex  $\sigma$  for all  $\sigma \in K$ .

Simplicial complexes are a good starting point, but they are insufficient for our purposes. A straightforward first generalization is as follows.

### 1.1.1 Definition (Abstract simplicial complexes)

An **abstract simplicial complex** is a collection of sets  $K = \{X^n\}_{n < \omega}$ , where  $X^k$  is a collection of subsets of  $X^0$  of cardinality  $k + 1$ . Furthermore, it must satisfy the condition that any subset of  $X^k$  of size  $j + 1$  belongs to  $X^j$ .

Elements of  $X^0$  are called the **vertices** of the complex.

Alternatively an abstract simplicial complex is a set  $K$  of finite subsets of  $X^0$  which is closed under taking subsets. Then  $X^n$  is just the collection of sets in  $K$  of cardinality  $n + 1$ .

In a simplicial complex of  $\mathbb{R}^N$  we required also that the intersection of simplices be a simplex, but this is because we were dealing with simplices embedded in an ambient space.

The standard  $n$ -simplex  $|\Delta^n|$  gives rise to the abstract simplicial complex of non-empty subsets of  $\{0, \dots, n\}$ .

### 1.1.2 Definition (Face maps)

We will assume that an abstract simplicial complex  $K$  comes equipped with a total ordering on its vertices. Then for an  $n$ -simplex  $\{v_0, \dots, v_n\} \in K$  we order its vertices and write  $[v_0, \dots, v_n]$  to mean  $v_0 < \dots < v_n$ . We further define the **face maps** of the simplicial complex  $K$  to be maps  $d_0, \dots, d_n: X^n \rightarrow X^{n-1}$  given by

$$d_i[v_0, \dots, v_n] = [v_0, \dots, \hat{v}_i, \dots, v_n]$$

where the hat means the vertex was removed.

For each  $n$  we have maps  $d_i$  for  $i \leq n$ , to differentiate the face maps between simplices of differing dimension we could write  $d_i^n$  instead of merely  $d_i$ , but this is an overload of information that is not necessary.

There is a fundamental relation between face maps which is immediately clear:

$$d_i d_j = d_{j-1} d_i, \quad \text{for } i < j. \tag{1}$$

Indeed, both sides map  $[v_0, \dots, v_n]$  to  $[v_0, \dots, \hat{v}_i, \dots, \hat{v}_j, \dots, v_n]$ .

**1.1.3 Definition (Simplicial maps)**

If  $K, L$  are simplicial complexes, a **simplicial map**  $f: K \rightarrow L$  is a function from vertices of  $K$  to vertices of  $L$  such that if  $\{v_0, \dots, v_n\} \in K$  then  $\{f(v_i)\}_i \in L$ . Note that a simplicial map need not be injective.

**1.2 Simplicial sets**

Simplicial sets, like simplicial complexes, can be given a combinatorial definition. Unfortunately the combinatorial definition is lengthier and instead has a more concise category-theoretic definition.

**1.2.1 Definition (Simplicial sets)**

The **simplex category**  $\Delta$  is the category whose objects are finite ordinals with order-preserving maps between them. Explicitly, its objects are  $[n] = \{0, \dots, n\}$  and a morphism  $f: [n] \rightarrow [m]$  is a function which preserves order (not necessarily strictly).

A **simplicial set** is a functor  $\Delta^{\text{op}} \rightarrow \mathbf{Set}$ . Then the category of simplicial sets is the functor category  $\mathbf{sSet} = [\Delta^{\text{op}}, \mathbf{Set}]$ . Morphisms are natural transformations.

Explicitly, a simplicial set  $X$  is a collection of sets  $X_n = X([n])$  along with functions  $f: X_n \rightarrow X_m$  for every order-preserving  $f: [m] \rightarrow [n]$ . Elements of  $X_0$  are called the **vertices** of the simplicial set.

To understand simplicial sets, it is worthwhile to understand the simplex category in more depth. There are some obvious maps in the simplex category  $\Delta$ :

- (1) The **coface map**  $\delta^i: [n-1] \rightarrow [n]$  (for  $i \leq n$ ) is the unique injection whose image leaves out only  $i$ . Explicitly,  $\delta^i(j) = j$  for  $j < i$  and  $\delta^i(j) = j+1$  for  $j \geq i$ .
- (2) The **codegeneracy map**  $\sigma^i: [n+1] \rightarrow [n]$  (for  $i \leq n$ ) is the unique surjection which picks out  $i$  twice. Explicitly,  $\sigma^i(j) = j$  for  $j \leq i$  and  $\sigma^i(j) = j-1$  for  $j > i$ .

The following identities are then easily verified:

$$\begin{aligned} \delta^j \circ \delta^i &= \delta^i \circ \delta^{j-1} \text{ for } i < j, \\ \sigma^j \circ \sigma^i &= \sigma^i \circ \sigma^{j+1} \text{ for } i \leq j, \\ \sigma^j \circ \delta^i &= \begin{cases} \delta^i \circ \sigma^{j-1} & i < j \\ \text{id} & i = j \text{ or } j+1 \\ \delta^{i-1} \circ \sigma^j & j+1 < i \end{cases}. \end{aligned}$$

Then if  $X$  is a simplicial set, we write  $X(\delta^i) = d_i: X_n \rightarrow X_{n-1}$  for the image of the coface maps, and  $X(\sigma^j) = s_j: X_n \rightarrow X_{n+1}$  for the image of the codegeneracy maps under  $X$ . These are called the **face** and **degeneracy** maps. Note that due to the contravariance of  $X$ , the face maps “shrink” and the degeneracy maps “grow”, while the reverse is true for the coface and codegeneracy maps.

The identities in the simplex category then dualize to

$$\begin{aligned} d_i d_j &= d_{j-1} d_i \text{ for } i < j, \\ s_i s_j &= s_{j+1} s_i \text{ for } i \leq j, \\ d_i s_j &= \begin{cases} s_{j-1} d_i & i < j \\ \text{id} & i = j \text{ or } j+1 \\ s_j d_{i-1} & j+1 < i \end{cases}. \end{aligned} \tag{1}$$

Note that the first identity is the same as the one in (1.1.1).

It is not hard to see that every morphism in the simplex category can be written as a composition of face and degeneracy maps. Thus we can equivalently define a simplicial set to be any collection of sets with face and degeneracy maps satisfying the identities in (1). And a morphism of simplicial sets  $X \rightarrow Y$  is a sequence of functions  $f_n: X_n \rightarrow Y_n$  which commute with the face and degeneracy maps:  $d \circ f = f \circ d$  and  $s \circ f = s \circ f$ .

If  $K$  is a simplicial complex, then it defines a simplicial set  $X$  where

$$\begin{aligned} X_n &= \{[v_0, \dots, v_n] \mid \text{The } v_i \text{ form a simplex in } K \text{ with } v_0 \leq \dots \leq v_n\}, \\ d_i[v_0, \dots, v_n] &= [v_0, \dots, \hat{v}_i, \dots, v_n], \\ s_i[v_0, \dots, v_n] &= [v_0, \dots, s_i, s_i, \dots, v_n]. \end{aligned}$$

Note that the elements of  $X_n$  can be *degenerate* simplices. For example if  $[0, 1, 2]$  is a simplex of  $K$ , then  $[0, 0, 1, 2, 2, 2]$  is a simplex of  $X$ . Essentially the simplicial set associated with a simplicial complex is obtained by adjoining all of these degenerate simplices.

We can generalize what we mean by a degenerate simplex.

### 1.2.2 Definition

Let  $X$  be a simplicial set. A **degenerate simplex** of  $X$  is a simplex  $x \in X$  which is in the image of some degeneracy map.

The prototypical simplicial set is

$$\Delta^n = \text{hom}_\Delta(\bullet, [n]).$$

We can better characterize  $\Delta^n$  as follows. Notice that an increasing map  $f: [k] \rightarrow [n]$  can be represented as a (degenerate)  $k$ -simplex  $[f(0), f(1), \dots, f(k)]$ . Thus we see that this definition of  $\Delta^n$  coincides with the one obtained from taking the simplicial complex defined by  $|\Delta^n|$  and adjoining degenerate simplices.

Notice by the Yoneda lemma, for a simplicial set  $X$ ,

$$\text{hom}_{\text{Set}}(\Delta^n, X) = \text{Nat}(\text{hom}_\Delta(\bullet, [n]), X) \cong X([n]) = X_n$$

where the isomorphism is obtained by mapping  $f: \Delta^n \rightarrow X$  to  $f(\text{id}_n)$  where  $\text{id}_n: [n] \rightarrow [n]$  is the identity.

### 1.2.3 Definition

For a simplicial set  $X$  and an  $n$ -simplex  $x \in X_n$ , we define  $\iota_x: \Delta^n \rightarrow X$  to be the unique simplicial map where  $\iota_x(\text{id}_n) = x$ . This is called the **representing map** of  $x$ .

## 1.3 Geometric realization

Consider the functor  $|\Delta^\bullet|: \Delta \rightarrow \mathbf{Top}$  which maps  $[n]$  to  $|\Delta^n|$ , and an order preserving map  $\varphi: [n] \rightarrow [m]$  to the continuous map  $|\Delta^n| \rightarrow |\Delta^m|$  given by

$$(x_0, \dots, x_n) \in |\Delta^n| \mapsto (y_0, \dots, y_m) \in |\Delta^m|, \quad y_j = \sum_{\varphi(i)=j} x_i.$$

For example, the face maps  $\delta^i$  define maps  $|\Delta^{n-1}| \rightarrow |\Delta^n|$  by

$$\delta^i(x_0, \dots, x_{n-1}) = (x_0, \dots, 0, \dots, x_{n-1}),$$

and the degeneracy maps  $\sigma^i: |\Delta^n| \rightarrow |\Delta^{n-1}|$  give

$$\sigma^i(x_0, \dots, x_n) = (x_0, \dots, x_i + x_{i+1}, \dots, x_n).$$

This then defines a functor

$$\text{hom}_{\mathbf{Top}}(|\Delta^\bullet|, \bullet): \Delta^{\text{op}} \times \mathbf{Top} \rightarrow \mathbf{Set}.$$

In particular for every topological space  $X$ , we have a simplicial set

$$\mathcal{S}(X) = \text{hom}_{\mathbf{Top}}(|\Delta^\bullet|, X),$$

this is called the **singular set** of  $X$ . The face and degeneracy maps of the singular set are then

$$d_i(\rho) = \rho \circ \delta^i, \quad s_i(\rho) = \rho \circ \sigma^i.$$

**Note:**

The singular set of  $X$  plays a central role in defining the singular homology of  $X$ . Recall that  $X$ 's chain complex is defined at each coordinate to be the free Abelian group generated by  $\mathcal{S}(X)_n$ :  $C_n(X) = Z[\mathcal{S}(X)_n]$ . The differentials are then defined in terms of the face maps:  $\partial: C_n(X) \rightarrow C_{n-1}(X)$  is defined on a singular  $n$ -simplex  $\sigma \in \mathcal{S}(X)_n$  by

$$\partial\sigma = \sum_{i=0}^n (-1)^i d_i \sigma,$$

and  $\partial$  is extended to all of  $C_n(X)$  by linearity.

Geometrically, we view a simplicial set  $X$  as a collection of (possibly degenerate)  $n$ -simplices  $X_n$  for all  $n < \omega$ . If we attach the geometric  $n$ -simplex  $|\Delta^n|$  to each  $n$ -simplex  $X_n$  we obtain  $X_n \times |\Delta^n|$ . We give each  $X_n$  the discrete topology, so this becomes a topological space. The geometric realization of  $X$  as a whole should be the union of all these simplices, but this is not enough. For one, degenerate simplices should not be counted; and we want to glue the faces of simplices together.

So we define the geometric realization as follows.

**1.3.1 Definition**

For a simplicial set  $X$ , we define its **geometric realization** to be the space

$$|X| = \coprod_{n < \omega} X_n \times |\Delta^n| / \sim$$

where each  $X_n$  has the discrete topology and  $\sim$  is the equivalence generated by the relations

$$(x, \delta^i p) \sim (d_i x, p), \quad (x, \sigma^i p) \sim (s_i x, p).$$

Here  $\delta^i: |\Delta^{n-1}| \rightarrow |\Delta^n|$  and  $\sigma^i: |\Delta^n| \rightarrow |\Delta^{n-1}|$  are the face and degeneracy maps respectively of the standard simplices.

For  $x \in X_n$ , we denote by  $(x, |\Delta^n|)$  the image of  $\{x\} \times |\Delta^n|$  in  $|X|$ .

**1.3.2 Lemma**

If  $X$  is a simplicial set, then every degenerate simplex of  $X$  is the degeneracy of a unique non-degenerate simplex. That is, if  $x \in X$  is degenerate then there is a unique non-degenerate  $y \in X$  such that  $x = s_{i_1} \cdots s_{i_k} y$  for some  $i_j$ s.

**Proof**

Clearly if  $x$  is degenerate there must exist *some*  $y$  which satisfies this. Now suppose  $x = Sy = S'y'$  for  $y, y'$  non-degenerate and  $S, S'$  a composition of degeneracy maps. Suppose  $S = s_{i_1} \cdots s_{i_k}$ , then define  $D = d_{i_k} \cdots d_{i_1}$ . This is a left inverse for  $S$ , and thus  $y = DS'y'$ .

Degeneracy and face maps commute according to the identities laid out in (1.2.1). Thus  $y = DS'y' = \tilde{S}\tilde{D}y'$  for some composition of degeneracy maps and face maps. Since  $y$  is by assumption non-degenerate,  $\tilde{S}$  must be empty so that  $y = \tilde{D}y'$ .

By symmetry,  $y' = \bar{D}y$  for some composition of face maps as well. Since face maps lower the dimension, this can only happen if  $\bar{D}, \tilde{D}$  are empty. Thus  $y = y'$ . ■

Similarly we have the following result.

**1.3.3 Lemma**

Every point  $v \in |\Delta^n|$  can be expressed uniquely as  $v = \delta^{i_1} \cdots \delta^{i_k} u$  where  $u$  is an interior point (all of its coordinates are positive).

Call a point  $(x, v) \in |X|$  **non-degenerate** if  $x$  is non-degenerate and  $v$  is interior.

**1.3.4 Lemma**

For a simplicial set  $X$ , every point in its geometric realization  $(x, v) \in |X|$  is equivalent to a unique non-degenerate point of  $|X|$ .

**Proof**

Define  $\lambda: |X| \rightarrow |X|$  as follows. For  $x \in X_n$  take  $i_1, \dots, i_k$  and  $y$  non-degenerate such that  $x = s_{i_1} \cdots s_{i_k} y$ . Then define

$$\lambda(x, v) = (y, \sigma^{i_k} \cdots \sigma^{i_1} v).$$

Similarly define  $\mu: |X| \rightarrow |X|$ , which for  $(x, v)$  maps

$$\mu(x, v) = (d_{i_k} \cdots d_{i_1} x, u)$$

where  $v = \delta^{i_1} \cdots \delta^{i_k} u$ .

Clearly  $\lambda, \mu$  both map points to equivalent points. Then  $\lambda \circ \mu$  maps a point to an equivalent non-degenerate point.

Uniqueness is because it can be verified that if  $p \sim p'$  then  $\lambda\mu(p) = \lambda\mu(p')$ . ■

An important fact is the following.

**1.3.5 Theorem**

The geometric realization of a simplicial set  $X$  is a CW complex.

**Proof**

Take the  $n$ -cells of  $|X|$  to be the images of the non-degenerate  $n$ -simplices of  $X$ . That is, simplices  $(x, |\Delta^n|)$  for  $x \in X_n$  non-degenerate. By the previous lemma, these cover  $|X|$ , and it can be verified that these make  $|X|$  into a CW complex. ■

**1.3.6 Theorem**

There is an adjunction  $|\bullet|: \mathbf{sSet} \rightleftarrows \mathbf{Top}: \mathcal{S}$  between the geometric realization and the singular set functors.

**Proof**

For a simplicial set  $X$  and a topological space  $Y$ , we define maps

$$\Phi: \mathbf{hom}_{\mathbf{Top}}(|X|, Y) \rightleftarrows \mathbf{hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y)): \Psi$$

as follows. For a continuous  $f: |X| \rightarrow Y$ , restricting  $f$  to a non-degenerate simplex  $(x, |\Delta^n|)$  restricts to a continuous map  $\Phi(f)_n: |\Delta^n| \rightarrow Y$ .

Conversely for a map  $g: X \rightarrow \mathcal{S}(Y)$ , for each  $n$ -simplex  $x \in X$  we have a continuous function  $\rho_x: |\Delta^n| \rightarrow Y$ . So define  $\Psi(g): |X| \rightarrow Y$  to act on the simplex  $(x, |\Delta^n|) \in |X|$  by applying  $\rho_x$  to  $|\Delta^n|$ . ■

## 1.4 Kan complexes

Recall that  $\mathbf{sSet} = [\Delta^{\text{op}}, \mathbf{Set}]$ . Since  $\mathbf{Set}$  is complete and cocomplete, so too is  $\mathbf{sSet}$ . That is to say,  $\mathbf{sSet}$  has all (small) limits and colimits. The co/limit of a functor is taken coordinate-wise, so such limits and colimits are easy to describe.

Most important for now is the product of simplicial sets. If  $K, L$  are simplicial sets, then their product  $K \times L$  is the simplicial set where

$$(K \times L)_n = K_n \times L_n \text{ for } n < \omega, \quad (K \times L)\varphi = (K\varphi) \times (L\varphi) \text{ for } \varphi: [n] \rightarrow [m].$$

Explicitly, the face and degeneracy maps are

$$d_i(x, y) = (d_i(x), d_i(y)), \quad s_i(x, y) = (s_i(x), s_i(y)).$$

If  $X$  is a simplicial set, a **simplicial subset** of  $X$  is a simplicial set  $Y$  such that  $Y_n \subseteq X_n$  for all  $n$  and the face and degeneracy maps of  $X$  restrict to those of  $Y$ . This is equivalent to  $Y$  being a subobject of  $X$  in the functor category  $\mathbf{sSet} = [\Delta^{\text{op}}, \mathbf{Set}]$  (see the following note).

### Note:

In a functor category  $[\mathcal{C}, \mathcal{D}]$  there are two notions of a “subfunctor”. Let  $F, G: \mathcal{C} \rightarrow \mathcal{D}$  be functors, then we may say that  $F$  is a subfunctor of  $G$  iff it is a subobject of  $G$  in the functor category. This means that there is a monic natural transformation  $F \Rightarrow G$ .

Another, generally easier to validate, definition is that  $F$  is a subfunctor of  $G$  iff  $F(c)$  is a subobject of  $G(c)$  in a natural way. That is to say, there is a natural transformation  $\alpha: F \Rightarrow G$  whose components  $\alpha_c: F(c) \rightarrow G(c)$  are monic.

In general clearly the latter implies the former: a pointwise monic natural transformation is clearly monic. Under relatively weak conditions, that  $\mathcal{D}$  has pullbacks, the former also implies the latter.

Recall that a morphism  $f: X \rightarrow Y$  is monic iff its pullback with itself is  $X$  (with projections given by the identity). That is, iff the following square is a pullback

$$\begin{array}{ccc} X & \xrightarrow{\quad \text{id} \quad} & X \\ \text{id} \downarrow & & \downarrow f \\ X & \xrightarrow{\quad f \quad} & Y \end{array}$$

Thus if  $\alpha: F \Rightarrow G$  is monic, then its pullback with itself is  $F$ . Limits in the functor category are computed pointwise, so this means that  $\alpha_c$ ’s pullback with itself is  $Fc$ . This means that  $\alpha_c$  is monic for all  $c$ , as desired.

Notice that if  $\{X^i\}_{i \in I}$  is a collection of simplicial subsets of  $X$ , then so is their intersection and union. This is because the  $X^i$  are invariant under any  $\varphi: X_m \rightarrow X_n$ , and thus so too is their intersection and union.

### 1.4.1 Definition

If  $S \subseteq X$  is an arbitrary subset (meaning a subset of  $\bigcup_n X_n$ ), then the simplicial subset it **generates** is the smallest simplicial subset of  $X$  containing  $S$  (meaning it contains  $S \cap X_n$  as  $n$ -simplices). Such a minimal simplicial subset exists as we can take the intersection of all simplicial subsets containing  $S$ .

For a morphism in the simplex category  $\varphi: [n] \rightarrow [m]$ , let us write  $X_\varphi: X_m \rightarrow X_n$  for the induced map. Then if  $S \subseteq X$  is an arbitrary set, then the simplicial subset it generates is just

$$(S) = \{X_\varphi(x) \mid x \in S\}.$$

This is a not-so-formal way of saying that  $(S)_n$  is the collection of all  $X_\varphi(x)$  where  $x \in S \cap X_k$  and  $\varphi: [n] \rightarrow [k]$ . Recall the standard  $n$ -simplex  $\Delta^n = \text{hom}_\Delta(\bullet, [n])$ . Its  $k$ -simplices  $\Delta_k^n$  are then order-preserving maps  $[k] \rightarrow [n]$ . In particular,  $\Delta_{n-1}^n$  consists only of  $\delta^0, \dots, \delta^n$ . Recall that the  $k$ -simplex in  $\Delta^n$  corresponding to  $\varphi: [k] \rightarrow [n]$  is given by  $[\varphi(0), \dots, \varphi(k)]$ . In particular  $\delta^i$  corresponds to  $[0, \dots, \hat{k}, \dots, n]$ .

#### 1.4.2 Definition

The **simplicial  $(n-1)$ -sphere**  $\partial\Delta^n$  is the simplicial subset of  $\Delta^n$  generated by  $\delta^0, \dots, \delta^n$ .

The **simplicial  $n$ -horn**  $\Lambda_k^n$  is the simplicial subset of  $\Delta^n$  generated by  $\delta^0, \dots, \delta^{\hat{k}}, \dots, \delta^n$  (all coface maps except for  $\delta^k$ ).

The  $i$ th face of  $\Delta^n$  is the simplicial subset  $\partial_i\Delta^n \subseteq \Delta^n$  generated by  $\delta^i$ . Since the union of simplicial subsets is a simplicial subset, we see that

$$\partial\Delta^n = \bigcup_{i \leq n} \partial_i\Delta^n, \quad \Lambda_k^n = \bigcup_{i \neq k} \partial_i\Delta^n.$$

Since  $\delta^k$  is the unique map without  $k$  in its image, geometrically we view  $\Lambda_k^n$  as the union of the faces of  $\Delta^n$  except for the one opposite the  $k$ th vertex.

Note that there exists a coequalizer diagram

$$\coprod_{0 \leq i < j \leq n} \Delta^{n-2} \rightrightarrows \coprod_{i=0}^n \Delta^{n-1} \rightarrow \partial\Delta^n$$

where the first morphisms have coordinates  $\Delta^{n-2} \rightarrow \coprod_i \Delta^{n-1}$  correspond to  $\delta^{j-1}$  and  $\delta^i$  respectively. That is,

$$\begin{array}{ccccc} \Delta^{n-2} & \xrightarrow{\delta^{j-1}} & \Delta^{n-1} & & \\ \text{inc}_{i < j} \downarrow & & \downarrow \text{inc}_i & \searrow \delta^i & \\ \coprod_{i < j} \Delta^{n-2} & \rightrightarrows & \coprod_i \Delta^{n-1} & \longrightarrow & \partial\Delta^n \\ \text{inc}_{i < j} \uparrow & & \uparrow \text{inc}_j & \nearrow \delta^j & \\ \Delta^{n-2} & \xrightarrow{\delta^i} & \Delta^{n-1} & & \end{array}$$

Similarly there is a coequalizer

$$\begin{array}{ccccc} \Delta^{n-2} & \xrightarrow{\delta^{j-1}} & \Delta^{n-1} & & \\ \text{inc}_{i < j} \downarrow & & \downarrow \text{inc}_i & \searrow \delta^i & \\ \coprod_{i < j} \Delta^{n-2} & \rightrightarrows & \coprod_{i \neq k} \Delta^{n-1} & \longrightarrow & \Lambda_k^n \\ \text{inc}_{i < j} \uparrow & & \uparrow \text{inc}_j & \nearrow \delta^j & \\ \Delta^{n-2} & \xrightarrow{\delta^i} & \Delta^{n-1} & & \end{array}$$

Using the universal property of coequalizers, the following lemma is immediate.

#### 1.4.3 Lemma

Let  $X$  be an arbitrary simplicial set.

- (1) A simplicial map  $\partial\Delta^n \rightarrow X$  is the same as a tuple  $(y_0, \dots, y_n)$  of  $(n-1)$ -simplices of  $X$  such that  $d_i y_j = d_{j-1} y_i$  for  $i < j$ .

- (2) A simplicial map  $\Lambda_k^n \rightarrow X$  is the same as a tuple  $(y_0, \dots, \hat{y}_k, \dots, y_n)$  of  $(n-1)$ -simplices of  $X$  such that  $d_i y_j = d_{j-1} y_i$  for  $i < j$ , neither equal to  $k$ .

### Proof

A map out of a coequalizer  $A \rightrightarrows B$  is just a map  $B \rightarrow X$  whose composition with either arrow in the coequalizer diagram is equal. Couple this with the fact that  $\text{hom}_{\text{Set}}(\Delta^n, X) \cong X_n$  and you have the desired result. ■

#### 1.4.4 Definition

Say that a simplicial set  $X$  has dimension  $\leq n$  if all  $k$ -simplices are degenerate for  $k > n$ . The  **$n$ -skeleton** of  $X$  is the largest simplicial subset of  $X$  with dimension  $\leq n$  (which exists as we can take unions). We denote it by  $\text{sk}_n(X)$ .

Alternatively, the  $n$ -skeleton of  $X$  is the simplicial subset generated by all non-degenerate simplices of dimension  $\leq n$ .

For example,  $\text{sk}_{n-1}(\Delta^n) = \partial\Delta^n$ . This is because  $\partial\Delta_k^n$  consists only of degenerate simplices for  $k \geq n$ , and for  $k < n$  we have  $\partial\Delta_k^n = \Delta_k^n$ .

In general we have a filtration of a simplicial set through its skeletons,

$$\emptyset = \text{sk}_{-1}(X) \subseteq \text{sk}_0(X) \subseteq \text{sk}_1(X) \subseteq \dots, \quad X = \bigcup_n \text{sk}_n(X).$$

#### 1.4.5 Definition

A simplicial map  $p: X \rightarrow Y$  is a **fibration** if for every commutative diagram of solid arrows, there is a dashed arrow making it commute

$$\begin{array}{ccc} \Lambda_k^n & \xrightarrow{\quad} & X \\ \downarrow & \nearrow \text{dashed} & \downarrow p \\ \Delta^n & \xrightarrow{\quad} & Y \end{array}$$

The arrow  $\Lambda_k^n \hookrightarrow \Delta^n$  is the inclusion map.

Fibrations are also called **Kan fibrations**.

Equivalently, for every tuple of  $(n-1)$ -simplices  $(x_0, \dots, \hat{x}_k, \dots, x_n)$  such that  $d_i x_j = d_{j-1} x_i$  for  $i < j$  not equal to  $k$ , if there is an  $n$ -simplex  $y \in Y_n$  such that  $d_i y = p(x_i)$ , then there is an  $n$ -simplex  $x \in X_n$  such that  $d_i x = x_i$  and  $p(x) = y$ .

There is a similar concept of fibration for topological spaces. A continuous map  $f: T \rightarrow U$  is a **Serre fibration** if a dashed arrow exists for any commutative diagram of continuous maps

$$\begin{array}{ccc} |\Lambda_k^n| & \xrightarrow{\quad} & T \\ \downarrow & \nearrow \text{dashed} & \downarrow f \\ |\Delta^n| & \xrightarrow{\quad} & U \end{array}$$

Because  $\mathcal{S}(|\Delta^n|) = \Delta^n$  and  $\mathcal{S}(|\Lambda_k^n|) = \Lambda_k^n$ , and geometric realization is left-adjoint to  $\mathcal{S}$ , such diagrams can be identified with diagrams

$$\begin{array}{ccc}
\Lambda_k^n & \xrightarrow{\quad} & \mathcal{S}(T) \\
\downarrow & \nearrow \text{dashed} & \downarrow \mathcal{S}(f) \\
\Delta^n & \xrightarrow{\quad} & \mathcal{S}(U)
\end{array}$$

So that  $f: T \rightarrow U$  is a Serre fibration iff  $\mathcal{S}(f): \mathcal{S}(T) \rightarrow \mathcal{S}(U)$  is a (Kan) fibration.

Notice that for any topological space  $X$  and one-point space  $*$ ,  $X \rightarrow *$  is a Serre fibration. This is because  $|\Lambda_k^n| \hookrightarrow |\Delta^n|$  is a strong deformation retract. Thus  $\mathcal{S}(X) \rightarrow \mathcal{S}(*)$  is a fibration of simplicial sets. Note that  $\mathcal{S}(*) = \Delta^0 = *$  is the terminal simplicial set (with a single simplex of each dimension).

This leads to the following definition.

#### 1.4.6 Definition

A simplicial set  $X$  is **fibrant**, or a **Kan complex**, if the map  $X \rightarrow *$  is fibrant.

Simplicial sets which arise from geometric simplicial complexes are not Kan complexes, thus  $\Delta^n$  is not. Neither is  $\partial\Delta^n$ .

The following lemma characterizes equivalent conditions for a simplicial set being fibrant, and is immediate.

#### 1.4.7 Lemma

Let  $X$  be a simplicial set, then the following are equivalent.

- (1)  $X$  is fibrant.
- (2) Every map  $f: \Lambda_k^n \rightarrow X$  extends to a map  $g: \Delta^n \rightarrow X$  for all  $n, k$ .

In a commutative diagram:

$$\begin{array}{ccc}
\Lambda_k^n & \xrightarrow{\quad} & X \\
\downarrow & \nearrow \text{dashed} & \\
\Delta^n & & 
\end{array}$$

- (3) For every tuple of  $(n-1)$ -simplices  $(x_0, \dots, \hat{x}_k, \dots, x_n)$  of  $X$  such that  $d_i x_j = d_{j-1} x_i$  for  $i < k$  not  $k$ , there is an  $n$ -simplex  $x \in X_n$  such that  $d_i x = x_i$ .

Fibrant simplicial sets are extremely important, as we will see soon.

An interesting result is that the underlying simplicial set of every simplicial group is fibrant. A simplicial group is a simplicial object in the category of groups, i.e. a functor  $\Delta^{\text{op}} \rightarrow \mathbf{Grp}$ . By underlying simplicial set, we mean the composition of the functor with the forgetful functor  $\mathbf{Grp} \rightarrow \mathbf{Set}$ .

#### 1.4.8 Lemma (Moore)

If  $H: \Delta^{\text{op}} \rightarrow \mathbf{Grp}$  is a simplicial group, its underlying set is fibrant.

## 1.5 Anodyne extensions

The homotopy theory of simplicial sets revolves around fibrations. Originally this was developed in terms of the extension condition, which results in tedious combinatorial manipulations. Fortunately with a more modern viewpoint we can suppress most of these combinatorial arguments.

**1.5.1 Definition**

A class  $M$  of monomorphisms of  $\mathbf{sSet}$  is called **saturated** if the following conditions are satisfied.

- (1)  $M$  contains all isomorphisms.
- (2)  $M$  is closed under pushouts in the sense that if

$$\begin{array}{ccc} A & \xrightarrow{\quad} & C \\ i \downarrow & & \downarrow i_* \\ B & \xrightarrow{\quad} & B \cup_A C \end{array}$$

is a pushout square and  $i \in M$ , then  $i_* \in M$  as well.

- (3) Each retract of an element of  $M$  is in  $M$ . That is, given the following commutative diagram

$$\begin{array}{ccccc} A' & \xrightarrow{\quad} & A & \xrightarrow{\quad} & A' \\ i' \downarrow & & \downarrow i & & \downarrow i' \\ B' & \xrightarrow{\quad} & B & \xrightarrow{\quad} & B' \end{array}$$

where the horizontal arrows compose to the identity. If  $i \in M$  then  $i' \in M$ .

- (4)  $M$  is closed under countable many compositions and arbitrary direct sums. This means that if there is a countable chain

$$A_1 \xrightarrow{i_1} A_2 \xrightarrow{i_2} \dots$$

with all  $i_j \in M$ , then the canonical map  $A_1 \rightarrow \operatorname{colim} A_i$  is in  $M$ .

Similarly if  $i_j: A_j \rightarrow B_j$  for  $j \in I$  are all  $M$ , then so is

$$\coprod_j i_j: \coprod_{j \in I} A_j \rightarrow \coprod_{j \in I} B_j.$$

Note that the maps in (2), (3), (4) may not obviously be monic. But these are all easily checked to be so. We do this for (2), denote  $f: A \rightarrow C$  so that  $B \cup_A C$  is the disjoint union  $B \amalg C$  where we identify  $f(a)$  and  $i(a)$ . The map  $i_*$  then maps  $c \in C$  to its equivalence class  $[c]$ . Notice that because  $i$  is monic, we cannot identify  $f(a), f(a')$  as that would mean  $i(a) = i(a')$  and thus  $a = a'$ . Thus no two distinct elements of  $C$  are identified and thus  $i_*$  is monic, as desired.

**1.5.2 Definition**

A map  $p: X \rightarrow Y$  is said to have the **right lifting property (RLP)** with respect to a class of morphisms  $M$  if in any solid arrow diagram

$$\begin{array}{ccc} A & \xrightarrow{\quad} & X \\ i \downarrow & \nearrow & \downarrow p \\ B & \xrightarrow{\quad} & Y \end{array}$$

where  $i \in M$ , a dashed arrow making the diagram commute exists.

Dually we define the **left lifting property**.

**1.5.3 Lemma**

The class of monomorphisms  $M_p$  having the left lifting property (LLP) with respect to a fixed map  $p: X \rightarrow Y$  is saturated.

**Proof**

We prove condition (2). Suppose we have a commutative diagram

$$\begin{array}{ccccc} A & \longrightarrow & C & \longrightarrow & X \\ \downarrow i & & \downarrow j & & \downarrow p \\ B & \longrightarrow & B \cup_A C & \longrightarrow & Y \end{array}$$

where the left square is a pushout and  $i \in M_p$ ; we wish to show  $j \in M_p$ . Because  $i \in M_p$ , there is a  $\theta: B \rightarrow X$  so that

$$\begin{array}{ccc} A & \longrightarrow & X \\ \downarrow i & \nearrow \theta & \downarrow p \\ B & \longrightarrow & Y \end{array}$$

commutes. But then this defines maps  $B \rightarrow X, C \rightarrow X$  which commute and thus a map  $\theta_*: B \cup_A C \rightarrow X$  which commutes with the necessary morphisms. Thus  $j \in M_p$  as desired. ■

Clearly the intersection of saturated classes is saturated. Thus for any class of monomorphisms  $B$ , we have a saturated class  $M_B$  of all saturated classes containing  $B$ . This is called the saturated class **generated** by  $B$ . Alternatively  $M_B$  is called the **saturation** of  $B$ .

Now consider the following classes of monomorphisms.

**B<sub>1</sub>** The set of all inclusions  $\Lambda_k^n \subseteq \Delta^n$  for  $0 \leq k \leq n, n > 0$ .

**B<sub>2</sub>** The set of all inclusions

$$(\Delta^1 \times \partial\Delta^n) \cup (\{e\} \times \Delta^n) \subseteq (\Delta^1 \times \Delta^n), \quad e = 0, 1$$

where  $\{e\}$  corresponds to the simplicial set consisting of the vertex  $e$  and all of its degeneracies.

**B<sub>3</sub>** The set of all inclusions

$$(\Delta^1 \times Y) \cup (\{e\} \times X) \subseteq (\Delta^1 \times X), \quad e = 0, 1$$

where  $Y \subseteq X$  is an inclusion.

**1.5.4 Proposition**

The classes **B<sub>1</sub>**, **B<sub>2</sub>**, and **B<sub>3</sub>** have the same saturation.

**Proof**

Long and tedious. ■

**1.5.5 Definition**

The saturation of  $\mathbf{B}_1$  (equivalently  $\mathbf{B}_2, \mathbf{B}_3$ ) is called the class of **anodyne extensions**.

**1.5.6 Corollary**

A fibration is a map  $p: X \rightarrow Y$  which has the right lifting property with respect to all anodyne extensions.

**Proof**

Because the class of monics for which  $p$  has the right lifting property is saturated, and contains  $\Lambda_k^n \subseteq \Delta^n$  by definition, we obtain the desired result. ■

**1.5.7 Corollary**

If  $i: K \hookrightarrow L$  is an anodyne extension, and  $Y \hookrightarrow X$  is an arbitrary inclusion, then

$$(K \times X) \cup (L \times Y) \rightarrow (L \times X)$$

is an anodyne extension.

**Proof**

Consider the class of monics  $K' \rightarrow L'$  such that

$$(K' \times X) \cup (L' \times Y) \rightarrow (L' \times X)$$

is anodyne. This is a saturated set.

Now consider the inclusion

$$(\Delta^1 \times Y') \cup (\{e\} \times X') \subseteq (\Delta^1 \times X'), \quad Y' \subseteq X'$$

We want to show that this is in the class, as it would imply all anodyne morphisms are. It is in the class iff the following inclusion is anodyne:

$$((\Delta^1 \times Y') \cup (\{e\} \times X')) \times X \cup (\Delta^1 \times X' \times Y) \rightarrow (\Delta^1 \times X' \times X)$$

which is just isomorphic to

$$(\Delta^1 \times ((Y' \times X) \cup (X' \times Y))) \cup (\{e\} \times X' \times X) \rightarrow \Delta^1 \times X' \times X$$

which is anodyne as desired.

Thus all anodyne extensions are in the class, in particular  $i$ . ■

**Note:**

In the above corollary, the simplicial set  $(K \times X) \cup (L \times Y)$  is the union when viewing both terms as subsets of  $L \times X$ . Categorically, this is the simplicial set  $(K \times X) \cup_{K \times Y} (L \times Y)$ .

**1.6 Function complexes**

In this section we will discuss some constructions in  $\mathbf{sSet}$ .

We first define quotient simplicial objects. First we define the quotient of a set by a subset: if  $A \subseteq B$  is a subset, then  $B/A = (B - A) \sqcup \{*\}$ ; we identify  $A$  with a single point and all other elements of  $B$  with themselves. We will also denote the additional point  $*$  by  $[A]$ .

Now if  $Y \subseteq X$  is a simplicial subset, then we can form the **quotient** simplicial set  $X/Y$  as follows. Then for  $n < \omega$  define

$$(X/Y)_n = X_n/Y_n,$$

and let  $q_n: X_n \rightarrow X_n/Y_n$  be the quotient map. Then for an order-preserving  $\varphi: [n] \rightarrow [m]$  (which we identify with  $X_m \rightarrow X_n$ ), there is a unique  $\bar{\varphi}: (X/Y)_m \rightarrow (X/Y)_n$  such that

$$\bar{\varphi} \circ q_m = q_n \circ \varphi$$

which maps  $x \in X_m - Y_m$  to  $q_n \circ \varphi(x)$  and  $y \in Y_m$  to  $[Y_n]$ . Uniqueness implies that  $\varphi \circ \bar{\psi} = \bar{\psi} \circ \bar{\varphi}$ , so that this is a functor. Thus  $X/Y$  is a simplicial set itself, and  $q: X \rightarrow X/Y$  is a morphism.

We can generalize this: let  $R \subseteq X \times X$  be a simplicial subset such that  $R_n \subseteq X_n \times X_n$  is an equivalence relation for all  $n < \omega$ . Then we can define the quotient  $X/R$  where the set of  $n$ -simplices is  $(X/R)_n = X_n/R_n$  (quotient by an equivalence relation). As before if we let  $q_n: X_n \rightarrow X_n/R_n$  be the quotient map, then for  $\varphi: [n] \rightarrow [m]$  there is a unique  $\bar{\varphi}: (X/R)_m \rightarrow (X/R)_n$  such that

$$\bar{\varphi} \circ q_m = q_n \circ \varphi.$$

This is well-defined precisely because  $R \subseteq X \times X$  so if  $q_m(x) = q_m(y)$ , meaning  $(x, y) \in R_m$ , then  $(\varphi(x), \varphi(y)) \in R_n$  and thus  $\bar{\varphi} \circ q_m(x) = q_n \circ \varphi(x)$  is well-defined. Thus  $X/R$  is a simplicial set and  $q: X \rightarrow X/R$  is a morphism. The quotient  $X/Y$  is a special case of quotienting by an equivalence. For each  $n < \omega$ ,  $Y_n$  defines an equivalence on  $X_n \times X_n$  where  $x_0 \sim x_1$  iff they are equal or both in  $Y_n$ . It is readily checked that this equivalence is simplicial, and then  $X/Y$  and  $X/R$  are equal.

A **pointed simplicial set** is a simplicial set  $X$  with a distinguished vertex  $* \in X_0$ . We denote by  $*$  also all of  $* \in X_0$ 's degeneracies. A morphism of pointed simplicial sets is a morphism of the underlying simplicial sets which preserves the base point. Note that it is sufficient to check that  $* \in X_0$  is mapped to  $* \in Y_0$ , as then naturality implies  $S*$  is mapped to  $S*$ . We denote the category of pointed simplicial sets by  $\mathbf{sSet}_*$ .

For two pointed simplicial sets  $X, Y$ , we define their **wedge sum**  $X \vee Y$  to be the simplicial subset of  $X \times Y$  consisting of all  $(x, *)$  and  $(*, y)$ . This is clearly a simplicial subset, and we define the **smash product** of  $X$  and  $Y$  to be the quotient  $X \wedge Y = X \times Y / X \vee Y$ . It is easy to see that the wedge sum  $X \vee Y$  is the pushout of the diagram  $X \leftarrow * \rightarrow Y$ , so the quotient of  $X \amalg Y$  by identifying  $* \in X$  with  $* \in Y$ .

If  $f: X \rightarrow Y$  is a simplicial map and  $K \subseteq Y$  is a simplicial subset, then  $f^{-1}K \subseteq X$  defined to be all the simplices of  $X$  which map to  $K$  via  $f$ , is a simplicial subset. Indeed, if  $\varphi: [m] \rightarrow [n]$  is a morphism, and  $x \in X_n$  maps to  $f(x) \in K_n$  then  $f(\varphi_*x) = \varphi_*f(x) \in K$  and so  $\varphi_*x \in f^{-1}K$  as desired.

We can view this fiber as the following pullback

$$\begin{array}{ccc} f^{-1}L & \longrightarrow & L \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

In particular for a vertex  $* \in Y$ , it generates the simplicial subset consisting of all its degeneracies, and the fiber of the vertex is  $f^{-1}*$  (the preimage of this simplicial subset).

Similarly we easily see that the image of a simplicial map is a simplicial subset of the codomain.

Now we focus on function complexes. First, consider the following functor  $\Delta^\bullet: \Delta \rightarrow \mathbf{sSet}$  given by  $[n] \mapsto \Delta^n$ . This is just the Yoneda embedding; it maps  $\varphi: [n] \rightarrow [m]$  to postcomposition with  $\varphi$ , which we identify with  $\varphi: \Delta^n \rightarrow \Delta^m$ . Thus we can view  $\Delta^\bullet$  as a simplicial object in  $\mathbf{sSet}$  (a simplicial simplicial set). Since taking products is also functorial, for any simplicial set  $X \in \mathbf{sSet}$ , this defines a functor

$$X \times \Delta^\bullet, \quad [n] \mapsto X \times \Delta^n, \quad \varphi \mapsto 1 \times \varphi_*.$$

For  $Y \in \mathbf{sSet}$ , we also have a functor  $\mathrm{hom}_{\mathbf{sSet}}(\bullet, Y): \mathbf{sSet}^{\mathrm{op}} \rightarrow \mathbf{Set}$ . Composing these functors we get a simplicial set  $\mathrm{hom}_{\mathbf{sSet}}(X \times \Delta^\bullet, Y): \Delta^{\mathrm{op}} \rightarrow \mathbf{Set}$ .

We spell this construction out explicitly.

**1.6.1 Definition (function complex)**

Let  $X$  and  $Y$  be simplicial sets. Their **function complex**  $\mathbf{Hom}(X, Y)$  to be the simplicial set whose simplices are

$$\mathbf{Hom}(X, Y)_n = \text{hom}_{\mathbf{sSet}}(X \times \Delta^n, Y)$$

and for  $\varphi: [m] \rightarrow [n]$ , it induces the map

$$\varphi^*: \mathbf{Hom}(X, Y)_n \rightarrow \mathbf{Hom}(X, Y)_m$$

defined by mapping  $f: X \times \Delta^n \rightarrow Y$  to the composite

$$X \times \Delta^m \xrightarrow{1 \times \varphi} X \times \Delta^n \xrightarrow{f} Y.$$

The construction of the function complex is functorial in both  $X$  and  $Y$ . That is the function complex is a functor

$$\mathbf{Hom}: \mathbf{sSet}^{\text{op}} \times \mathbf{sSet} \rightarrow \mathbf{sSet}.$$

A morphism  $f: X \rightarrow X'$  defines  $f^*: \mathbf{Hom}(X', Y) \rightarrow \mathbf{Hom}(X, Y)$  given by precomposition with  $f \times \text{id}$ . That is, it maps  $g: X' \times \Delta^n \rightarrow Y$  to

$$f^*(g) = X \times \Delta^n \xrightarrow{f \times \text{id}} X' \times \Delta^n \xrightarrow{g} Y = g \circ (f \times \text{id}).$$

Similarly  $f: Y \rightarrow Y'$  defines  $f_*: \mathbf{Hom}(X, Y) \rightarrow \mathbf{Hom}(X, Y')$  by postcomposition with  $f$ . So  $g: X \times \Delta^n \rightarrow Y$  is mapped to  $f \circ g$ .

There is a natural **evaluation map**

$$\text{ev}: X \times \mathbf{Hom}(X, Y) \rightarrow Y, \quad (x, f) \mapsto f(x, \text{id}_n)$$

where  $x \in X_n$  and  $f: X \times \Delta^n \rightarrow Y$ . For a simplicial structure map  $\varphi: [m] \rightarrow [n]$  we have that

$$\text{ev}(\varphi(x, f)) = \text{ev}(\varphi x, \varphi f) = \varphi f(\varphi x, \text{id}_n) = f(\varphi x, \varphi(\text{id})) = \varphi f(x, \text{id}) = \varphi \text{ev}(x, f).$$

Thus  $\text{ev}$  is a simplicial map.

**1.6.2 Proposition (exponential law)**

The function

$$\text{ev}_*: \text{hom}_{\mathbf{sSet}}(K, \mathbf{Hom}(X, Y)) \rightarrow \text{hom}_{\mathbf{sSet}}(X \times K, Y)$$

defined by mapping  $g: K \rightarrow \mathbf{Hom}(X, Y)$  to the composite

$$X \times K \xrightarrow{1 \times g} X \times \mathbf{Hom}(X, Y) \xrightarrow{\text{ev}} Y$$

is a bijection natural in  $K, X, Y$ .

**Proof**

The inverse of  $\text{ev}_*$  is the map

$$\text{hom}_{\mathbf{sSet}}(X \times K, Y) \rightarrow \text{hom}_{\mathbf{sSet}}(K, \mathbf{Hom}(X, Y))$$

obtained by mapping  $g: X \times K \rightarrow Y$  to  $g_*: K \rightarrow \mathbf{Hom}(X, Y)$  where for  $x \in K_n$ ,

$$g_*(x) = X \times \Delta^n \xrightarrow{1 \times \iota_x} X \times K \xrightarrow{g} Y.$$

Indeed,  $\text{ev}_*(g_*)$  maps  $(x, k) \in X_n \times K_n$  to

$$(x, k) \mapsto \text{ev}(x, g_*(k)) = g_*(k)(x, \text{id}_n) = g(x, \iota_k \text{id}_n) = g(x, k)$$

so that  $\text{ev}_*(g_*) = g$ . And for  $g: K \rightarrow \mathbf{Hom}(X, Y)$ ,  $\text{ev}_*(g)_*$  maps  $k \in K_n$  to  $\text{ev}_*(g)_*(k) \in \text{hom}_{\mathbf{sSet}}(X \times \Delta^n, Y)$  defined by

$$(x, \varphi) \mapsto \text{ev}_*(g)(x, \iota_k \varphi) = \text{ev}(x, g(\iota_k \varphi)) = g(\iota_k \varphi)(x, \text{id}_n)$$

where  $\varphi \in \Delta^n$ . But  $\iota_k(\varphi) = \varphi(k)$  and so this is equal to

$$= g(\varphi(k))(x, \text{id}_n) = (\varphi g(k))(x, \text{id}_n)$$

and recall that in the definition of  $\mathbf{Hom}(X, Y)$ ,  $\varphi$  acts on it by

$$= g(k)(x, \varphi(\text{id}_n)) = g(k)(x, \varphi)$$

so that  $\text{ev}_*(g)_* = g$  as desired. ■

Thus  $\mathbf{sSet}$  is **cartesian closed**: its product has a left adjoint. The function complex  $\mathbf{Hom}$  is also called the **interior hom**.

**Note:**

In general for a category  $\mathcal{C}$ , its presheaf category  $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$  is cartesian closed. The internal hom of presheaves is given much like here. For presheaves  $F, G: \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$  we define the presheaf

$$\mathbf{Hom}(F, G): \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}, \quad \mathbf{Hom}(F, G)(c) = \text{Nat}(F \times y(c), G)$$

where  $y: \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$  is the Yoneda embedding.

### 1.6.3 Proposition

If  $i: K \hookrightarrow L$  is an inclusion of simplicial sets and  $p: X \rightarrow Y$  is a fibration, then the map

$$\mathbf{Hom}(L, X) \xrightarrow{(i^*, p_*)} \mathbf{Hom}(K, X) \times_{\mathbf{Hom}(K, Y)} \mathbf{Hom}(L, Y)$$

which is induced by the diagram

$$\begin{array}{ccc} \mathbf{Hom}(L, X) & \xrightarrow{p_*} & \mathbf{Hom}(L, Y) \\ \downarrow i^* & & \downarrow i^* \\ \mathbf{Hom}(K, X) & \xrightarrow{p_*} & \mathbf{Hom}(K, Y) \end{array}$$

is a fibration.

**Proof**

Diagrams of the form

$$\begin{array}{ccccc}
 \Lambda_k^n & \xrightarrow{\quad} & \mathbf{Hom}(L, X) & & \\
 \downarrow & & \downarrow (i^*, p_*) & & \\
 \Delta^n & \xrightarrow{\quad} & \mathbf{Hom}(K, X) \times_{\mathbf{Hom}(K, Y)} \mathbf{Hom}(L, Y) & \xrightarrow{\quad} & \mathbf{Hom}(K, X) \\
 & & \downarrow & & \downarrow \\
 & & \mathbf{Hom}(L, Y) & \xrightarrow{\quad} & \mathbf{Hom}(K, Y)
 \end{array}$$

correspond to, according to the exponential law, diagrams of the form

$$\begin{array}{ccccc}
 \Lambda_k^n \times K & \xrightarrow{\quad} & \Lambda_k^n \times L & & \\
 \downarrow & & \downarrow & & \\
 \Delta^n \times K & \xrightarrow{\quad} & (\Lambda_k^n \times L) \cup_{\Lambda_k^n \times K} (\Delta^n \times K) & \xrightarrow{\quad} & X \\
 & & \downarrow j & & \downarrow p \\
 & & \Delta^n \times L & \xrightarrow{\quad} & Y
 \end{array}$$

By corollary 1.5.7  $j$  is anodyne. Since  $p$  is fibrant, it has the right lifting property relative to  $j$  and thus the desired lift exists in the lower diagram; this corresponds to a lift in the original diagram. ■

**1.6.4 Corollary**

- (1) If  $p: X \rightarrow Y$  is a fibration, so is  $p_*: \mathbf{Hom}(K, X) \rightarrow \mathbf{Hom}(K, Y)$ .
- (2) If  $i: K \hookrightarrow L$  is an inclusion and  $X$  is fibrant, then  $i^*: \mathbf{Hom}(L, X) \rightarrow \mathbf{Hom}(K, X)$  is a fibration.

**Proof**

- (1) Consider the inclusion  $i: * \hookrightarrow K$  and the isomorphism

$$\mathbf{Hom}(K, Y) \cong * \times_* \mathbf{Hom}(K, Y) \cong \mathbf{Hom}(*, X) \times_{\mathbf{Hom}(*, Y)} \mathbf{Hom}(K, Y).$$

Under this isomorphism,  $p_*$  corresponds to  $(i^*, p_*)$  which is fibrant by the above proposition.

- (2) Consider the map  $p: X \rightarrow *$  which is fibrant by definition. Then as in the previous point,

$$\mathbf{Hom}(K, X) \cong \mathbf{Hom}(K, X) \times_{\mathbf{Hom}(K, *)} \mathbf{Hom}(L, *)$$

under which  $i$  corresponds to  $(i^*, p_*)$  which is fibrant by the above proposition. ■

**1.7 Simplicial homotopy****1.7.1 Definition**

Let  $f, g: K \rightarrow X$  be simplicial maps. A **simplicial homotopy** from  $f$  to  $g$  is a map  $h: K \times \Delta^1 \rightarrow X$  such that the following diagram commutes

$$\begin{array}{ccc}
 K \times \Delta^0 = K & & \\
 \downarrow 1 \times \delta^1 & \searrow f & \\
 K \times \Delta^1 & \xrightarrow{h} & X \\
 \uparrow 1 \times \delta^0 & \nearrow g & \\
 K \times \Delta^0 = K & & 
 \end{array}$$

If there exists a simplicial homotopy from  $f$  to  $g$  we write  $f \simeq g$ .

Recall that a  $k$ -simplex in  $\Delta^n$  is an increasing  $f: [k] \rightarrow [n]$  which we can represent as  $[f(0), \dots, f(k)]$ . In particular its vertices can naturally be written as  $0, \dots, n$  according to the image of  $f: [0] \rightarrow [n]$ .

Note that  $\delta^1: \Delta^0 \rightarrow \Delta^1$  corresponds to the vertex of  $\Delta^1$  given by  $\delta^1: [0] \rightarrow [1]$ , which by definition skips 1; its image is 0. Similarly  $\delta^0$  corresponds to the vertex 1.

So a homotopy can equivalently be given as a simplicial map  $h: K \times \Delta^1 \rightarrow X$  such that

$$h(x, 0) = f(x), \quad h(x, 1) = g(x)$$

for all  $x \in K$  (where  $0, 1$  denote the degeneracies of the vertices  $0, 1$ ).

### 1.7.2 Definition

Suppose  $\iota: L \hookrightarrow K$  is an inclusion of simplicial sets, and let  $f, g: K \rightarrow X$  be simplicial maps. Further suppose that  $f|_L = g|_L$ . A homotopy from  $f$  to  $g$  **relative to**  $L$  is a homotopy  $h: f \simeq g$  such that the following diagram commutes.

$$\begin{array}{ccc}
 K \times \Delta^1 & \xrightarrow{h} & X \\
 \uparrow \iota \times \text{id} & & \uparrow f|_L = g|_L \\
 L \times \Delta^1 & \xrightarrow{\text{proj}_L} & L
 \end{array}$$

This means that  $h(x, t) = f(x) = g(x)$  for all  $x \in L, t \in \Delta^1$ . A homotopy of the form

$$L \times \Delta^1 \xrightarrow{\text{proj}_L} L \xrightarrow{\alpha} X$$

is called a constant homotopy: it is a homotopy from  $\alpha$  to itself.

#### Note:

Homotopy is not in general an equivalence. Consider the maps  $\iota_0, \iota_1: \Delta^0 \rightarrow \Delta^n$  classifying the vertices  $0, 1$  respectively. A homotopy between them is a map  $h: \Delta^1 \rightarrow \Delta^n$  such that  $h(0) = 0$  and  $h(1) = 1$ . This corresponds to the 1-simplex  $[0, 1] \in \Delta^n$ .

But there is not a homotopy from  $\iota_1$  to  $\iota_0$  as this would correspond to a simplex  $[1, 0]$  which does not exist.

For an  $n$ -simplex  $\sigma \in X$ , denote its **boundary** by

$$\partial\sigma = (d_0\sigma, \dots, d_n\sigma).$$

**1.7.3 Lemma**

Suppose that  $X$  is a fibrant simplicial set. Then the simplicial homotopy of vertices  $x: \Delta^0 \rightarrow X$  is an equivalence.

**Proof**

There is a homotopy  $x \rightsquigarrow y$  iff there is a 1-simplex  $v \in X_1$  such that  $d_1v = x$  and  $d_0v = y$ . That is, iff there is a 1-simplex  $v$  with boundary  $\partial v = (y, x)$ . Note that  $\partial(s_0x) = (x, x)$  gives reflexivity.

Now suppose  $\partial v_2 = (y, x)$  and  $\partial v_0 = (z, y)$  so that  $d_0v_2 = d_1v_0 = y$ . Thus  $v_2, v_0$  determine a map  $(v_0, \cdot, v_2): \Lambda_1^2 \rightarrow X$ . Because  $X$  is fibrant, we can lift it,

$$\begin{array}{ccc} \Lambda_1^2 & \xrightarrow{(v_0, \cdot, v_2)} & X \\ \downarrow & \nearrow \theta & \\ \Delta^2 & & \end{array}$$

Then

$$\partial(d_1\theta) = (d_0d_1\theta, d_1d_1\theta) = (d_0d_0\theta, d_1d_2\theta),$$

and

$$d_0\theta = \theta \circ \delta^0 = (v_0, \cdot, v_2) \circ \delta^0 = v_0$$

and similar for  $d_2\theta$ . Thus

$$\partial(d_1\theta) = (d_0v_0, d_1v_2) = (z, x)$$

so that  $d_1\theta$  is a homotopy  $x \rightsquigarrow z$ ; so homotopy is transitive.

For symmetry, suppose  $\partial v_2 = (y, x)$  and let  $v_1 = s_0x$ . Then  $d_1v_1 = d_1v_2 = x$ , so these define a map  $(\cdot, v_1, v_2): \Lambda_0^2 \rightarrow X$ . We extend this to  $\theta': \Delta^1 \rightarrow X$  so that

$$\partial(d_0\theta') = (d_0d_0\theta', d_1d_0\theta') = (d_0d_1\theta', d_0d_2\theta') = (x, y).$$

Thus the relation is symmetric. ■

**1.7.4 Lemma**

If  $f: X \rightarrow Y$  is a fibration and  $y \in Y$  is a vertex, then  $f^{-1}y \subseteq X$  is a fibrant simplicial set.

**Proof**

Consider the diagram

$$\begin{array}{ccccc} \Lambda_k^n & \longrightarrow & f^{-1}y & \longrightarrow & X \\ \downarrow & & \downarrow & & \downarrow f \\ \Delta^n & \longrightarrow & * & \longrightarrow & Y \end{array}$$

where  $* \rightarrow Y$  chooses the vertex  $y$ . Because  $f$  is fibrant, there is a lifting  $\theta: \Delta^n \rightarrow X$ . But the bottom arrow is the constant map to  $y$ , and so  $f \circ \theta = y$  which means  $\theta$  maps into  $f^{-1}y$ . Thus  $\theta$  is really a lift  $\Delta^n \rightarrow f^{-1}y$ , as desired. ■

**1.7.5 Corollary**

If  $X$  is fibrant and  $L \subseteq K$  is an inclusion, then

- (1) homotopy of maps  $K \rightarrow X$  is an equivalence, and
- (2) homotopy of maps  $K \rightarrow X \pmod{L}$  is an equivalence.

### Proof

Clearly (2) implies (1) with  $L = \emptyset$ . Now we know that  $i^*: \mathbf{Hom}(L, X) \rightarrow \mathbf{Hom}(K, X)$  is a fibration, and the fiber of a vertex  $\alpha \in \mathbf{Hom}(K, X)_0$  consists of all  $f \in \mathbf{Hom}(L, X)$  such that  $f|_L = \alpha$  (recall  $\mathbf{Hom}(K, X)_0 = \text{hom}_{\text{Set}}(k, X)$ ). These fibers are all fibrant by the above lemma.

Now we claim that  $f \xrightarrow{\sim} g \pmod{L}$  iff  $f, g$  are homotopic as vertices of the fiber of  $i^*$ . Let  $\alpha = f|_L = g|_L$ . A homotopy  $h: h \xrightarrow{\sim} g \pmod{L}$  corresponds to a map  $h: \Delta^1 \times K \rightarrow X$ , which by the exponential law corresponds to  $\tilde{h}: \Delta^1 \rightarrow \mathbf{Hom}(K, X)$ .

Note that a homotopy diagram of the form

$$\begin{array}{ccc}
 K \times \Delta^0 = K & & \\
 \downarrow 1 \times \delta^1 & \searrow f & \\
 K \times \Delta^1 & \xrightarrow{h} & X \\
 \uparrow 1 \times \delta^0 & \nearrow g & \\
 K \times \Delta^0 = K & & 
 \end{array}$$

Corresponds to a homotopy of vertices

$$\begin{array}{ccc}
 \Delta^0 & & \\
 \delta^1 \downarrow & \searrow \iota_f & \\
 \Delta^1 & \xrightarrow{\tilde{h}} & \mathbf{Hom}(K, X) \\
 \delta^0 \uparrow & \nearrow \iota_g & \\
 \Delta^0 & & 
 \end{array}$$

and vice versa.

Homotopy relative to  $L$  is the same as homotopy of vertices in the fibers of  $i^*$ , which are fibrant. By the above lemma this is an equivalence.  $\blacksquare$

## 1.8 Simplicial homotopy groups

**1.8.1 Definition**

Let  $X$  be a fibrant simplicial set and  $v \in X_0$  a vertex. For  $n \geq 1$ , define  $\pi_n(X, v)$  to be the set of homotopy classes of maps  $\alpha: \Delta^n \rightarrow X$  (rel  $\partial\Delta^n$ ) such that  $\alpha$  fits into a diagram

$$\begin{array}{ccc} \Delta^n & \xrightarrow{\alpha} & X \\ \uparrow & & \uparrow v \\ \partial\Delta^n & \xrightarrow{\quad} & * \end{array}$$

Equivalently,  $\alpha$  restricts to the constant map  $v$  on  $\partial\Delta^n$ .

For  $n = 0$  define  $\pi_0(X)$  to be the set of homotopy classes of vertices of  $X$ . This is called the **path components** of  $X$ .  $X$  is said to be **connected** if  $\pi_0(X)$  is trivial (a singleton).

We write  $[\alpha]$  for the homotopy class of  $\alpha$ .

Note that  $\alpha$  restricting to  $v$  on  $\partial\Delta^n$  just means that  $d_i\alpha = v$  for all  $i$ . Indeed,  $d_i\alpha$  is just  $\alpha \circ \delta^i$  and since  $\delta^i$  is in  $\partial\Delta^n$ , this is mapped to  $v$ .

Let  $\alpha, \beta: \Delta^n \rightarrow X$  represent elements of  $\pi_n(X, v)$ . Then define the simplices

$$v_i = v \text{ for } 0 \leq i \leq n-2, \quad v_{n-1} = \alpha, \quad v_{n+1} = \beta.$$

Then  $d_i v_j = d_{j-1} v_i$  for  $i < j$  and  $i, j \neq n$ , as all the faces are just  $v$ . Thus the  $v_i$  define a simplicial map  $(v_0, \dots, v_{n-1}, v_n): \Lambda_n^{n+1} \rightarrow X$  and as  $X$  is fibrant there is an extension

$$\begin{array}{ccc} \Lambda_n^{n+1} & \xrightarrow{(v_0, \dots, v_{n-1}, v_n)} & X \\ \downarrow & \nearrow \omega & \\ \Delta^{n+1} & & \end{array}$$

Notice then that

$$\partial(d_n \omega) = (d_0 d_n \omega, \dots, d_{n-1} d_n \omega, d_n d_n \omega) = (d_{n-1} d_0 \omega, \dots, d_{n-1} d_{n-1} \omega, d_n d_{n+1} \omega) = (v, \dots, v).$$

Thus  $d_n \omega$  represents an element of  $\pi_n(X, v)$ .

**1.8.2 Lemma**

The homotopy class of  $d_n \omega$  (rel  $\partial\Delta^n$ ) is independent on choice of representatives of  $[\alpha]$  and  $[\beta]$  and choice of extension  $\omega$ .

**Proof**

Suppose that  $h_{n-1}: \alpha \xrightarrow{\sim} \alpha'$  (rel  $\partial\Delta^n$ ) and  $h_{n+1}: \beta \xrightarrow{\sim} \beta'$  (rel  $\partial\Delta^n$ ) are homotopies and

$$\partial\omega = (v, \dots, v, \alpha, d_n \omega, \beta), \quad \partial\omega' = (v, \dots, v, \alpha', d_n \omega', \beta')$$

are extensions.

There is a map

$$(\Delta^{n+1} \times \partial\Delta^n) \cup (\Lambda_n^{n+1} \times \Delta^1) \xrightarrow{(\omega, \omega', (v, \dots, v, h_{n-1}, h_{n+1}))} X$$

given by this data. We can then extend this to  $\theta: \Delta^{n+1} \times \Delta^1$ . The composite

$$\Delta^n \times \Delta^1 \xrightarrow{\delta^n \times 1} \Delta^{n+1} \times \Delta^1 \xrightarrow{\theta} X$$

is a homotopy  $d_n \omega \xrightarrow{\sim} d_n \omega'$  (rel  $\partial \Delta^n$ ). ■

## TODO: What?

From this lemma, we can define multiplication on  $\pi_n(X, v)$  by  $[\alpha] \cdot [\beta] = [d_n \omega]$ , where  $\partial \omega = (v, \dots, v, \alpha, d_n \omega, \beta)$ . Let  $e \in \pi_n(X, v)$  denote the homotopy class of  $v$ , i.e. the constant map  $\Delta^n \rightarrow * \xrightarrow{v} X$ .

### 1.8.3 Theorem

For a simplicial set  $X$  and a vertex  $v \in X_0$ ,  $\pi_n(X, v)$  forms a group for all  $n \geq 1$ .

#### Proof

First we show that  $\alpha \cdot e = \alpha$ . To do this, it is sufficient to find an  $\omega: \Delta^{n+1} \rightarrow X$  such that  $d_{n-1} \omega = d_n \omega = \alpha$  and  $d_i \omega = v$  for all other  $i$ . Then  $\partial \omega = (v, \dots, v, \alpha, \alpha, v)$  showing  $\alpha \cdot e = \alpha$ . We can take  $\omega = s_{n-1} \alpha$  and this is sufficient.

Now we want to show that  $\alpha$  has an inverse, so there is a  $\beta$  where  $\alpha \cdot \beta = e$ . We want an  $\omega$  with  $d_n \omega = v$ ,  $d_{n-1} \omega = \alpha$ , and  $d_i \omega = v$  for all other  $i$ . That is, we want to complete  $(v, \dots, v, \alpha, v, )$ . This is possible as  $X$  is fibrant.

Now we show associativity. Let  $\alpha, \beta, \gamma$  represent elements of  $\pi_n(X, v)$ . Then we have  $(n+1)$ -simplices  $\omega_{n-1}, \omega_{n+1}, \omega_{n+2}$  representing  $\alpha\beta$ ,  $(\alpha\beta)\gamma$ , and  $\beta\gamma$  respectively. That is,

$$\begin{aligned} \partial \omega_{n-1} &= (v, \dots, v, \alpha, d_n \omega_{n-1}, \beta), \\ \partial \omega_{n+1} &= (v, \dots, v, d_n \omega_{n-1}, d_n \omega_{n+1}, \gamma), \\ \partial \omega_{n+2} &= (v, \dots, v, \beta, d_n \omega_{n+2}, \gamma). \end{aligned}$$

These form a map

$$\Lambda_n^{n+2} \xrightarrow{(v, \dots, v, \omega_{n-1}, \omega_{n+1}, \omega_{n+2})} X$$

which extends to a map  $\theta: \Delta^{n+2} \rightarrow X$ . But then

$$\partial(d_n \theta) = (v, \dots, v, \alpha, d_n \omega_{n+1}, d_n \omega_{n+2})$$

so that  $\alpha \cdot d_n \omega_{n+2} = d_n \omega_{n+1}$ .

Unrolling definitions, we have

$$(\alpha\beta)\gamma = d_n \omega_{n-1} \cdot \gamma = d_n \omega_{n+1} = \alpha \cdot d_n \omega_{n+2} = \alpha(\beta\gamma)$$

as desired. ■

We will show that  $\pi_n(X, v)$  is an Abelian group for  $n \geq 2$ . To do this, we will construct the *loop space*  $\Omega X$  where  $\pi_n(X, v) \cong \pi_{n-1}(\Omega X, v)$  and show that  $\pi_k(\Omega X, v)$  is Abelian for all  $k \geq 1$ .

#### Note:

We digress a bit here for the sake of some background. The constructions thus far all have a topological analogue, which you should be familiar with. The loop space does as well, which we describe here.

First of all, let us construct the **function space** of topological spaces. Let  $X$  and  $Y$  be topological spaces, then their function space is the space  $C(X, Y)$  of continuous functions  $X \rightarrow Y$ . We define a subbase as

follows: for  $K \subseteq X$  compact and  $U \subseteq Y$  open, define

$$V(K, U) = \{f \in C(X, Y) \mid f(K) \subseteq U\}.$$

These define our subbase as  $K, U$  range. This is called the **compact-open topology**.

An important result is that, if  $X$  is locally compact Hausdorff, then  $C(X, Y)$  forms an internal hom. That is, there is an adjunction  $\bullet \times X \dashv C(X, \bullet)$ . The adjunction is given by currying: map  $\alpha: Y \times X \rightarrow Z$  to  $\tilde{\alpha}: Y \rightarrow C(X, Z)$  given by  $\tilde{\alpha}(y)(x) = \alpha(y, x)$ . And similarly map  $\beta: Y \rightarrow C(X, Z)$  to  $\hat{\beta}: Y \times X \rightarrow Z$  by  $\hat{\beta}(y, x) = \beta(y)(x)$ . It is standard to see that these are well-defined (i.e.  $\tilde{\alpha}, \hat{\beta}$  are continuous); this is dependent on  $X$  being locally compact Hausdorff.

The adjunction defines natural isomorphisms

$$C(Y \times X, Z) \cong C(Y, C(X, Z)),$$

and it is again standard to show that these isomorphisms are continuous. Thus  $C(Y \times X, Z)$  is homeomorphic to  $C(Y, C(X, Z))$ . We have then enriched **Top** over itself, and shown that the internal hom is an internal hom in this enrichment.

We are a bit more concerned with  $\mathbf{Top}_*$ , the category of pointed topological spaces (i.e. topological spaces with a distinguished point). If  $X, Y$  are pointed topological spaces, then  $C_*(X, Y)$  is the subspace of  $C(X, Y)$  consisting of basepoint-preserving maps  $X \rightarrow Y$ . On the other hand,  $X \times Y$  has no natural basepoint: any choice of  $X \times *$  or  $* \times Y$  would be valid. Thus we define their **smash product** to be  $X \wedge Y = (X \times Y)/(X \times * \cup * \times Y)$ . It is then standard to show that the above adjunction restricts to

$$C_*(Y \wedge X, Z) \cong C_*(Y, C_*(X, Z)).$$

We now define the **loop space** of  $X$  to be  $\Omega X = C_*(S^1, X)$ , for any choice of basepoint of  $S^1$ . Similarly we define the **suspension** of  $X$  to be  $\Sigma X = X \wedge S^1$ . The adjunction then shows that

$$C_*(\Sigma X, Y) \cong C_*(X, \Omega Y).$$

Now we claim that

$$\pi_0 C(X, Y) \cong [X, Y]$$

where the right-hand side denotes the collection of maps  $X \rightarrow Y$  quotiented by homotopy. Indeed, a path  $I \rightarrow C(X, Y)$  corresponds under the adjunction to a homotopy  $I \times X \rightarrow Y$  and vice versa. The endpoints of the path are the edges of the homotopy, giving the desired result. Similarly if we let  $[X, Y]_*$  denote the class of basepoint-preserving homotopies, i.e. maps  $h: I \times X \rightarrow Y$  such that  $h(t, *) = *$ , then

$$\pi_0 C_*(X, Y) \cong [X, Y]_*.$$

Now we know that by definition,  $\pi_1 X = [S^1, X]_*$ . Thus

$$\pi_1 X \cong \pi_0 C_*(S^1, X) = \pi_0 \Omega(X),$$

as desired. By induction this means that  $\pi_0 \Omega^n(X) \cong \pi_n(X)$ , so  $\pi_{n+1} X \cong \pi_n \Omega(X)$ .

Further note that

$$[\Sigma X, Y]_* \cong \pi_0 C_*(\Sigma X, Y) \cong \pi_0 C_*(X, \Omega Y) \cong [X, \Omega Y]_*.$$

This is called **Eckmann-Hilton duality**.